Writing Findings & Feature Engineering

Fixing the two most common data science mindset gaps

The Real Problem: Thinking Like a Programmer, Not a Domain Expert

When you look at column names and 'nothing clicks,' and when you can spot observations but can't say what they imply — both happen because you're looking at data as rows and columns rather than as a story about real-world things. This is the single most important mindset shift in data science.

Problem 1: Writing Findings — You're Missing the 'So What'

Your findings sound vague because you stop at the observation. Every finding has three parts:

What — what you observed

e.g. "Feature X is right-skewed"

Why it matters — what it implies

e.g. "meaning most values are low with a few extreme highs"

What you'll do about it — the action

e.g. "I'll apply a log transform before modeling"

You're writing only the first part. Practice forcing yourself to always complete the other two. Use this sentence structure every time:

> *"[Observation], which means [implication], so I will [action]."*

It feels mechanical at first but it trains the habit. **Vocabulary isn't your real problem — structure is.** Once you have structure, the right words follow naturally.

Problem 2: Feature Engineering — You're Skipping the Domain Thinking Step

When nothing clicks from column names, it's because you're jumping straight to 'what technique do I apply?' instead of asking 'what is this data actually about?' Before touching any code, spend 5 minutes asking:

- What real-world process generated this data?
- What would a human expert in this field care about?
- What combinations or ratios would be meaningful to a person, not a model?

A concrete example:

> If you have an e-commerce dataset with purchase_amount and num_items — a programmer sees two columns. A domain thinker sees **average item price** (purchase_amount / num_items), which might be far more predictive than either column alone. The feature came from thinking about shopping behavior, not from thinking about data.

This is why practicing on datasets from familiar domains helps enormously. If you've ever shopped online, taken a loan, or tracked fitness — you already have domain intuition you're not using.

What To Do This Week

For writing:

> Take your last EDA and rewrite every finding using the What → Why it matters → Action structure. Do this for 5 findings. You'll immediately feel the difference.

For feature engineering:

> Next dataset you open, close your laptop for 10 minutes first and just think — 'what is this data about, and what would a smart person in this field want to measure that isn't directly in the columns?' Write 3 ideas on paper before coding anything. Even if the ideas are wrong, this habit is what separates people who can engineer features from those who can't.

The gap you're experiencing isn't about vocabulary or techniques. It's about building the habit of thinking about **meaning before jumping to code.**